

0.

```
void main (void){
    int m=func(7,5)
    int L=m+3;
    return 0;
}
int func(int a, int c){
    int m=10;
    m=m+a+c
    return m;
}
```

main:

```
addi a0, zero, 7
addi a1, zero, 5
jal x1, func
addi t0, a0, 3
ret
```

func:

```
addi sp, sp, -4
sw s0, 0(sp)

addi s0, zero, 10
add s0, s0, a0
add s0, s0, a1
mv a0, s0

lw s0, 0(sp)
addi sp, sp, 4
ret
```

1. Write a snippet of assembly code that implements the following:

$$x9 = x6 + x7 - 31 + x8$$

2. Write a snippet of assembly code that multiplies the value in register x11 by 5

3. Write an assembly program that implements the following code:

```
if ( x < z)
    y = x
Else
    y = z + 10
```

```
#x8 = x,   x9 = y,   x10 = z
```

4. Design an assembly subroutine to compute the factorial function: $\text{fact}(n) = n!$

```
fact=1
for(i=1;i<=n;i++)
    fact=fact*i
```

5. Suppose that register x5 holds the status of some LEDs. Write a snippet of assembly code that turns off the lower 8 LEDs. The upper 8 LEDs should be unchanged. (i.e. the least significant 8 LEDs should be turned off and others should be unchanged).

```
#0x11080000 = memory mapped address for LEDs
```

6. Write the equivalent assembly code for the C function below.

```
extern int f2 (int, int);
extern int f3 (int *, int);
int f1 (int a, int b) {
    return f2 (a, f3(&a, b));
}
```

7. Write the equivalent assembly code for the C function below.

```
int A[20];
int sum = 0;
for (int i=0; i<20; i++)
    sum += A[i];
sol:
```